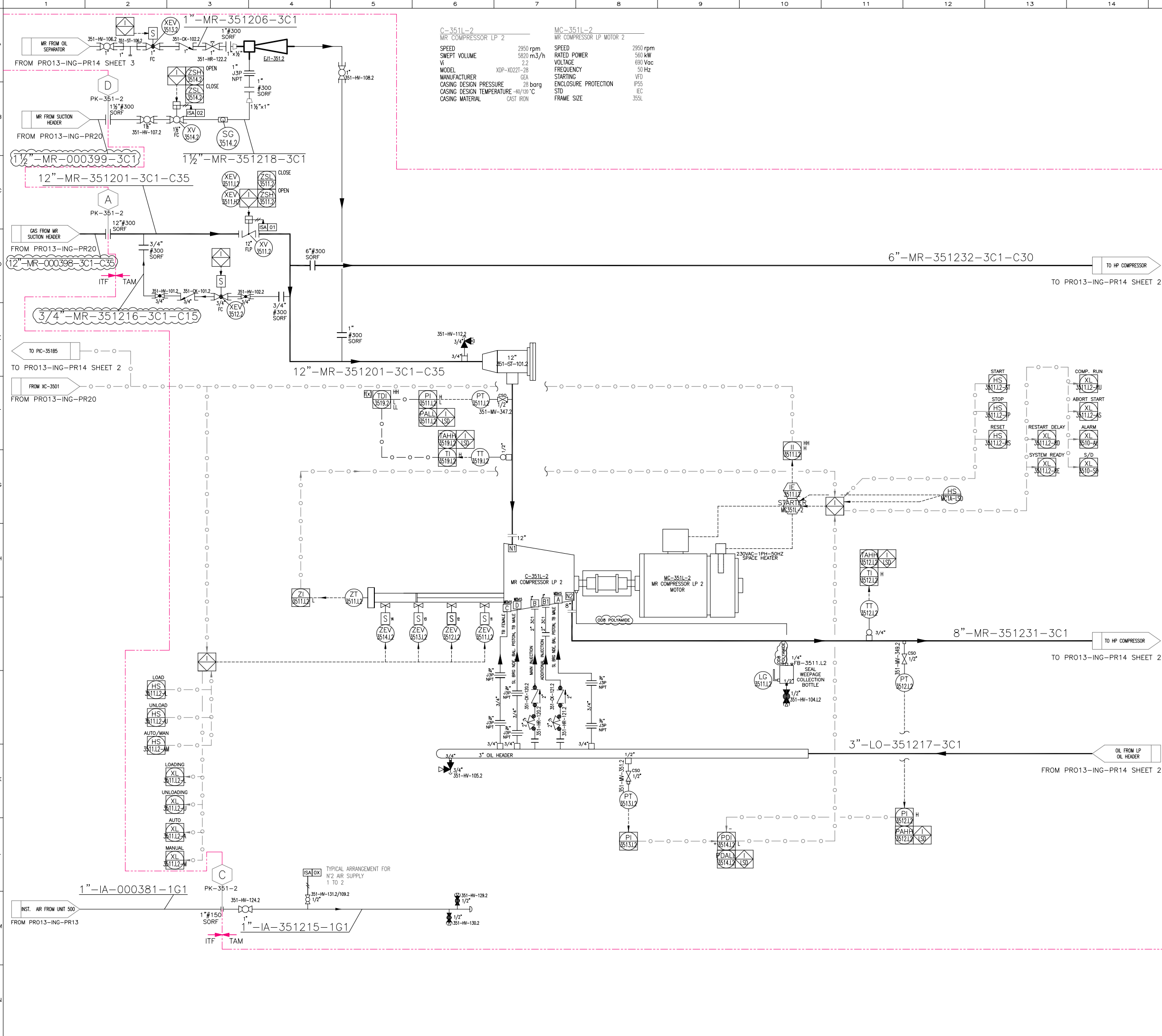




REFERENCE DOCUMENTS

P&ID LEGEND	PRO13-ING-PRO0
PROCESS FLOW DIAGRAM	21044-PR-PID-00000
UTILITY FLOW DIAGRAM	PRO10-ING-PRO1/02/03/04
MATERIAL SELECTION DIAGRAM	21044-PR-PFD-00001
	PRO10-ING-PR11
	21044-PR-UFD-00001
	PRO08-ING-FR01
	21044-PR-MSD-00001

INSTRUMENT SUPPLY AIR

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NOTES:

----- TAM SUPPLY LIMIT

1	25 Mar 2026	ISSUE FOR COMMISSIONING	TAM (JB)	TAM (MM)	LCAMPLONE
0	08 Jan 2026	AS BUILT	TAM (JB)	LCAMPLONE	W.BOUALALI
Rev.	Date	Description	Draft	Check'd	Approved

Customer: MANAENERGY

Contractor: ITALFLUID

Engineering: ITALFLUID

Technology Vendor: TECHNICAL AMERICA INC

Offices: Mahaj Riad Center, Batiment 6, Jeme elage, Avenue Al Az, Hay Ryad, 10100 Rabat, Morocco

Offices: VIA VVALDI,4 MONTESILVANO (ITALY) Tel.+39 085 4224374 fax+39 085 4220742

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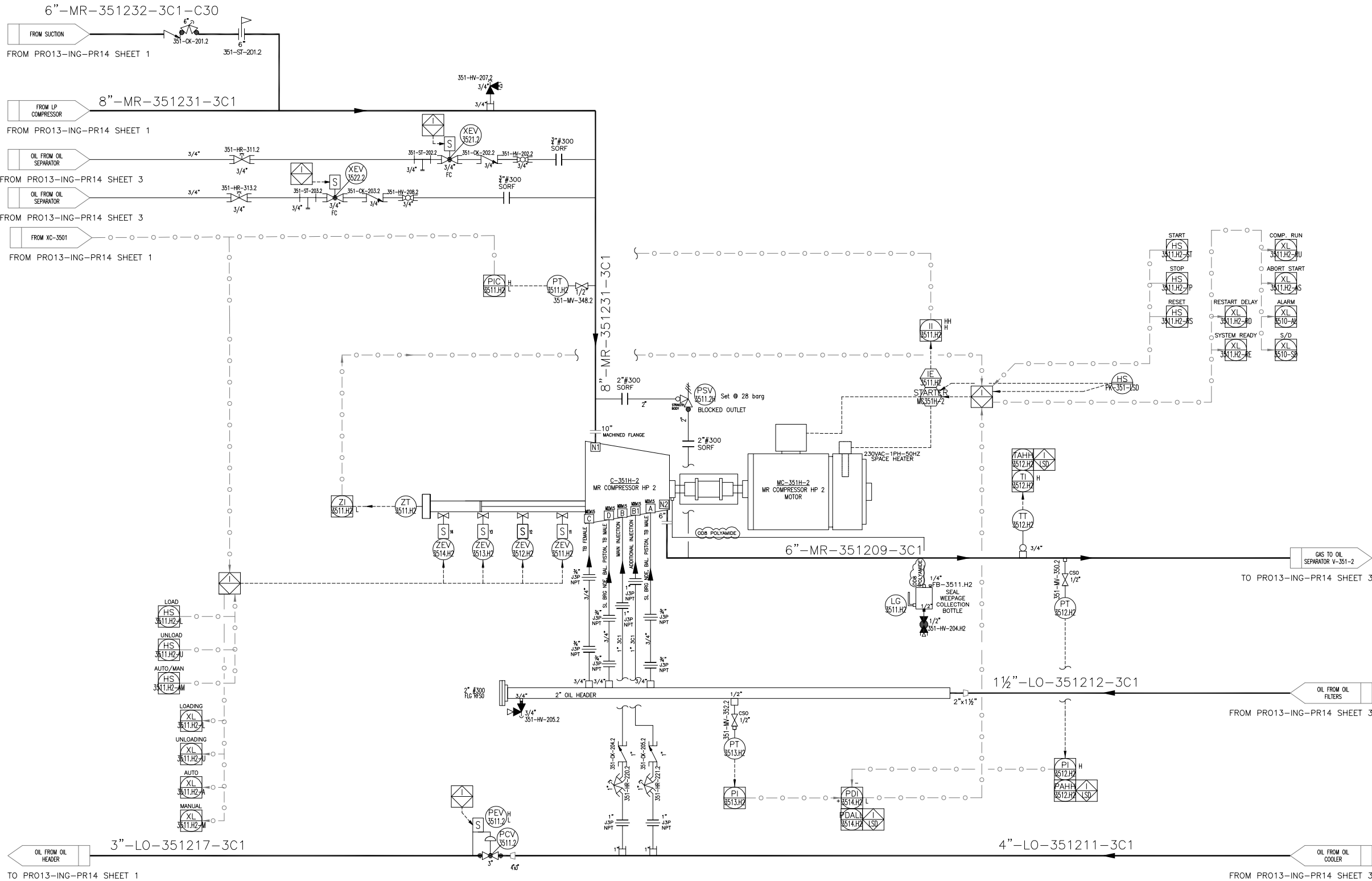
Offices: 301 N Smith Avenue, 92880 Corona, California (USA) Ph.+1 951 272 9540

TITLE: LNG TENDRARA PLANT

OBJECT: P&ID PACKAGE PK-351-2 (UNIT 350)

TYPE	SCALE	DOCUMENTS ID:	VENDOR DOC CODE:	SHEET/OF
A1	N/A	PRO13-ING-PR14	21044-PR-PID-35002	1/3

C-351H-2		MC-351H-2	
MR COMPRESSOR HP 2		MR COMPRESSOR HP MOTOR 2	
SPEED	2950 rpm	SPEED	2950 rpm
SWEEP VOLUME	1997 m3/h	RATED POWER	560 kW
VI	2.2	VOLTAGE	690 Vdc
MODEL	WP-W36S-28	FREQUENCY	50 Hz
MANUFACTURER	GEA	STARTING	VFD
CASING DESIGN PRESSURE	28 barg	ENCLOSURE PROTECTION	IP55
CASING DESIGN TEMPERATURE	-60/120 °C	STD	IEC
CASING MATERIAL	CAST IRON	FRAME SIZE	355L



REFERENCE DOCUMENTS

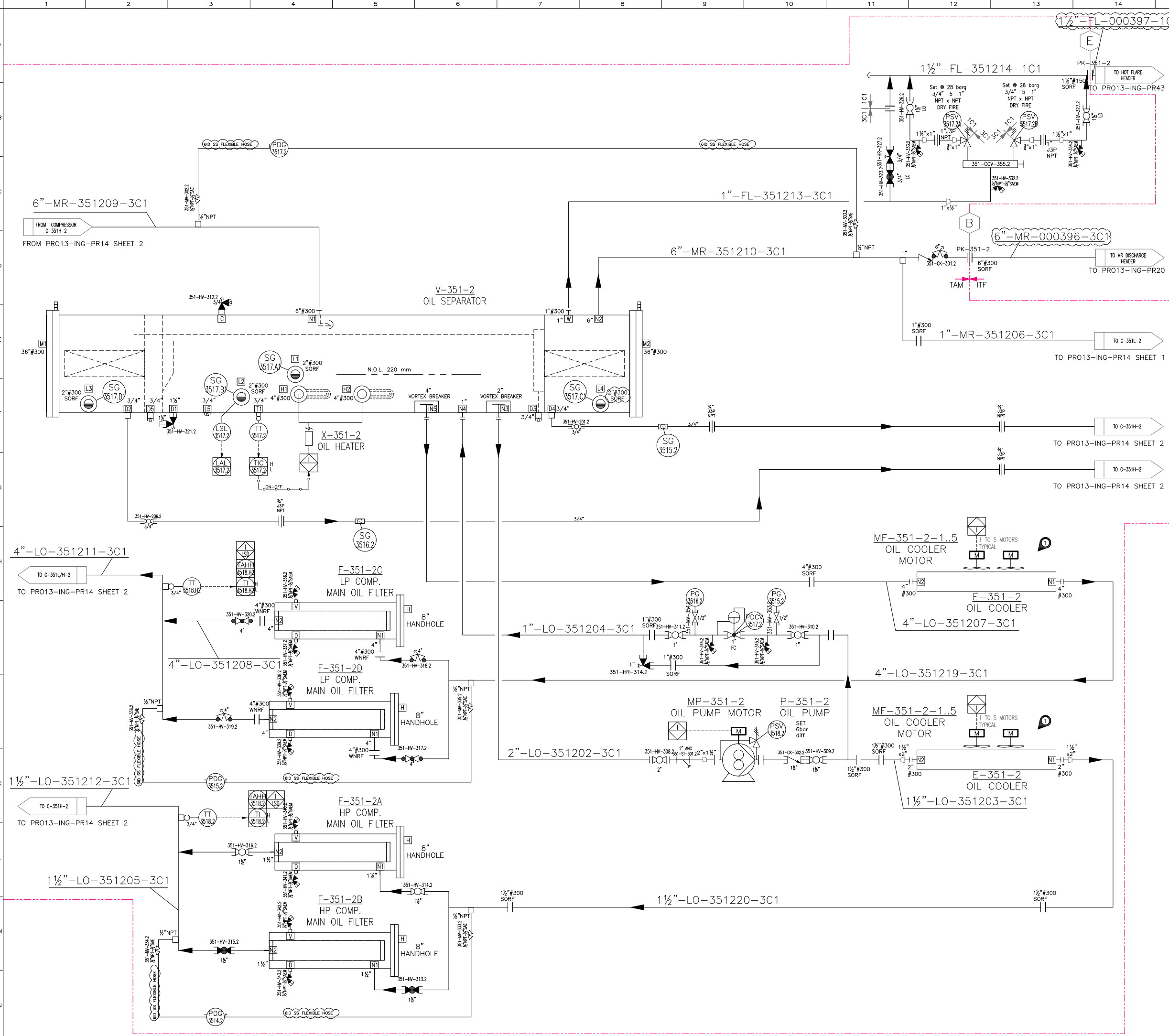
P&ID LEGEND	PRO13-ING-PRO0
PROCESS FLOW DIAGRAM	21044-PR-PID-00000
UTILITY FLOW DIAGRAM	PRO10-ING-PRO1/02/03/04
MATERIAL SELECTION DIAGRAM	21044-PR-PFD-00001
	PRO10-ING-PR11
	21044-PR-UFD-00001
	PRO08-ING-FR01
	21044-PR-MSD-00001

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Rev.	Date	Description	Draft	Check'd	Approved
Customer: MANAENERGY			OFFICES: Mahaj Riad Center, Batiment 6, 3eme etage, Avenue Al Az, Hay Ryad, 10100 Rabat, Morocco		
Contractor: ITALFLUID			OFFICES: VIA VIVALDI,4 MONTESILVANO (ITALY) Tel.+39 085 4224374 fax+39 085 4220742		
Engineering: ITALFLUID			OFFICES: VIA VIVALDI,4 MONTESILVANO (ITALY) Tel.+39 085 4224374 fax+39 085 4220742		
Technology Vendor: TECHNICAL AMERICA INC			OFFICES: 301 N Smith Avenue, 92880 Corona, California (USA) Ph.+1 951 272 9540		
TITLE: LNG TENDRARA PLANT			OBJECT: P&ID PACKAGE PK-351-2 (UNIT 350)		
TYPE	SCALE	DOCUMENTS ID:	VENDOR DOC CODE:		SHEET/OF
A1	N/A	PRO13-ING-PR14	21044-PR-PID-35002		2/3



REFERENCE DOCUMENTS

P&ID LEGEND	PRO13-ING-PROO
PROCESS FLOW DIAGRAM	21044-PR-PID-00000
UTILITY FLOW DIAGRAM	PRO10-ING-PRO1/02/03/04
MATERIAL SELECTION DIAGRAM	21044-PR-PFD-00001
	PRO10-ING-PR11
	21044-PR-UFD-00001
	PRO08-ING-FR01
	21044-PR-MSD-00001

V-351-2

OIL SEPARATOR

DESIGN PRESSURE	28 bar g
DESIGN TEMPERATURE	-15/120 °C
DIAMETER	965 00 mm
SM TO SM	5000 mm
CORROSION ALLOWANCE	0 mm
INSULATION	NO
LINING/CLADDING	N.A.
MATERIAL	CS
INTERNAL VOLUME	3.4 m ³
INTERNAL ELEMENTS	COALESCE

MP-351-2

OIL PUMP ELECTRIC MOTOR

SPEED	1450 rpm
RATED POWER	4 kW
VOLTAGE	690 Vac
FREQUENCY	50 Hz
STARTING	DOL
ENCLOSURE PROTECTION	IP55
STD	IEC
FRAME SIZE	112

F-351-2A/B

HP COMP. MAIN OIL FILTER

DESIGN PRESSURE	34 bar g
DESIGN TEMPERATURE	-15/120 °C
DIAMETER	8" NPS
SM TO SM	600 mm
CORROSION ALLOWANCE	0 mm
INSULATION	NO
LINING/CLADDING	N.A.
MATERIAL	CS
INTERNAL VOLUME	0.01 m ³
INTERNAL ELEMENTS	PARTICULATE FILTER

MF-351-2-1...10

OIL COOLER FAN ELECTRIC MOTORS

SPEED	1450 rpm
RATED POWER	4 kW
VOLTAGE	400/3 Vac
FREQUENCY	50 Hz
STARTING	DOL
ENCLOSURE PROTECTION	IP55
STD	IEC
FRAME SIZE	112M

F-351-2-1...10

FAN

SPEED	1450 rpm
RATED CAPACITY	27634 m ³ /h
ABSORBED POWER	3 kW

P-351-2

OIL PUMP

SPEED	1450 rpm
RATED CAPACITY	110 l/min
HEAD	4 bar
TYPE	VOLUMETRIC
MANUFACTURER	KRACHT
CASING DESIGN PRESSURE	34 barg
CASING DESIGN TEMPERATURE	-15/120 °C
MODEL	KF80
MOTOR POWER	4 kW
CASING MATERIAL	CI

X-351-2

OIL HEATER

RATED POWER	2.25 (3x 0.75) kW
SUPPLY VOLTAGE	690 Vac-50Hz-3ph

F-351-2

OIL COOLER

DESIGN DUTY	380 kW
DESIGN PRESSURE	28 / 34 bar g
DESIGN TEMP.	-15/120 °C
MATERIAL	AS304 / A1 in
CORR. ALL.	0 mm
FAN Qty	10
FAN ABS POWER each	3 kW
SURFACE AREA	4304 m ²

F-351-2C/D

LP COMP. MAIN OIL FILTER

DESIGN PRESSURE	28 bar g
DESIGN TEMPERATURE	-15/120 °C
DIAMETER	8" NPS
SM TO SM	1000 mm
CORROSION ALLOWANCE	0 mm
INSULATION	NO
LINING/CLADDING	N.A.
MATERIAL	CS
INTERNAL VOLUME	0.033 m ³
INTERNAL ELEMENTS	PARTICULATE FILTER

NOTES:

OIL COOLER CAN BE SPLIT IN 2 SEPARATE OIL COOLERS OR COMBINED IN ONE OIL COOLER WITH SEPARATE CIRCUIT.

Rev.

Date

Description

Customer:

MANAENERGY

Contractor:

ITALFLUID

Engineering:

ITALFLUID

Technology Vendor:

TECHNICAL AMERICA INC

OFFICES:

Manajir Rad Center, Batiment 6, 3eme etage, Avenue Al Az, Hay Ryad, 10100 Rabat, Morocco

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TITLE:

LNG TENDRARA PLANT

OBJECT:

P&ID PACKAGE PK-351-2 (UNIT 350)

TYPE

SCALE

DOCUMENTS ID:

PRO13-ING-PR14

VENDOR DOC CODE:

21044-PR-PID-35002

SHEET/OF

3/3

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